

Professional Preparation

University of California, Berkeley, Earth Science, PhD	2004
University of Arizona, Tucson, Geophysics, MS	1995
University of Montana, Missoula, BA	1992

Appointments

2020-present	Professor	Rice University
2022, 2026	Distinguished Visiting Professor	University of Hawaii, Manoa
2015-2020	Associate Professor	Rice University
2009-2015	Assistant Professor	Rice University
2006-2008	SOEST Young Investigator	University of Hawaii, Manoa
2005-2006	Daly Postdoctoral Fellow	Harvard University
1996-1999	Hydrogeologist	Geomatrix Consultants

Publications (Rice student/postdoctoral scholar in bold font)

Park, J.G., Gonnermann, H.M., Fliedner, C., Giachetti, T., Ryan, A.G. Bubble nucleation as a necessary condition for explosive magma fragmentation, *Earth Planet. Sci. Lett.* (in review, EPSL-D-26-00316).

Ferrante, G., Gonnermann, H.M. Bottoms up: Coupling versus decoupling within Kīlauea's magma supply system, *J. Volcanol. Geotherm. Res.*, 473, <https://doi.org/10.1016/j.jvolgeores.2026.108584> (2026).

Ferrante, G., Gonnermann, H.M., Fliedner, C., Giachetti, T., Ryan, A.G. Viscosity of bubbly magmas from torsional experiments on pumice, *J. Volcanol. Geotherm. Res.*, 461, 108297, <https://doi.org/10.1016/j.jvolgeores.2025.108297> (2025).

Barth, A., Portnyagin, M., Mironov, N., Holtz, F., Moussallam, Y., Rose-Koga, E.F., Rasmussen, D., Towbin, H., Gonnermann, H.M., Mutch, E.J.F., Rotolo, S.G., Plank, T. Fast, furious, and gassy: Etna's explosive eruption from the mantle, *Earth Planet. Sci. Lett.*, 643, <https://doi.org/10.1016/j.epsl.2024.118864> (2024).

Jones, T.J., Shetty, A., Chalk, C., Dufek, J., Gonnermann, H.M. Identifying rheological regimes within pyroclastic density currents, *Nature Comm.*, <https://doi.org/10.1038/s41467-024-48612-7> (2024).

Phelps, P., Gonnermann, H.M., Ruprecht, P., Winslow, H., Delgado, F., Pritchard, M. E., Liao, Y., Magma chamber replenishment from a crystal mush at Cordón Caulle, Chile, *Geophys. J. Int.* 235, 610-634, <https://doi.org/10.1093/gji/ggad259> (2023).

Winslow, H., Ruprecht, P., Gonnermann H. M., **Phelps, P.**, Muñoz-Saez, C., Delgado, F., Pritchard, M. E. and Amigo, A., Insights for crystal mush storage utilizing mafic enclaves from the 2011-12 Cordón Caulle eruption, *Scientific Reports*, 19:9734, <https://doi.org/10.1038/s41598-022-13305-y> (2022).

Crozier, J., Tramontano, S., Forte, P., Oliva, S. J. C., Gonnermann, H. M., Lev, E., Manga, M., Rader, E., Ruprecht, P., Myers, M., Paisley, R., Houghton, B. F., Shea, T., Amigo, A., Tuffen, H., Castro, J., Schipper, C. I., Outgassing through fractures enables effusive eruption of silicic magma, *J. Volcanol. Geotherm. Res.*, 430, 107617, <https://doi.org/10.1016/j.jvolgeores.2022.107617> (2022).

Hajimirza, S., **Jones, T. J.**, Gonnermann, H. M., Moreland, W., & Thordarson, T., Quantifying water entrainment in volcanic jets, *Geophys. Geochem. Geosys.*, 23, e2021GC010160. <https://doi.org/10.1029/2021GC010160> (2022).

Hajimirza, S., Gonnermann, H. M. & Gardner, J. E., Experimental demonstration of continuous bubble nucleation in rhyolite, *J. Volcanol. Geotherm. Res.*, 421, doi.org/10.1016/j.jvolgeores.2021.107417 (2022).

Carn, S. A., P. A. Newman, V. Aquila, H. Gonnermann, and J. Dufek, Anticipating climate impacts of major volcanic eruptions, *Eos*, 102, <https://doi.org/10.1029/2021EO162730> (2021).

Gonnermann, H. M. & Anderson, K., Modeling Volcano-Magmatic Systems: Workshop Report for the Modeling Collaboratory for Subduction Research Coordination Network, <https://doi.org/10.31223/X55G96> (2021).

Hajimirza, S., Gonnermann, H. M. & Gardner, J. E., Reconciling bubble nucleation in explosive eruptions with geospeedometers, *Nature Comms.*, 283, doi.org/10.1038/s41467-020-20541-1 (2021).

Barth, A., Newcombe, M., Plank, T., Gonnermann, H. M., **Hajimirza, S.**, Soto, G., Saballos, A. and Hauri, E., Magma decompression rate correlates with explosivity at basaltic volcanoes – Constraints from water diffusion in olivine. *J. Volcanol. Geotherm. Res.*, 387, doi.org/10.1016/j.jvolgeores.2019.106664, (2019).

Hajimirza, S., Gonnermann, H. M., Gardner, J. E., and Giachetti, T., Predicting homogeneous bubble nucleation in rhyolite. *J. Geophys. Res.*, 124, <https://doi.org/10.1029/2018JB015891> (2019).

Giachetti, T., Gonnermann, H.M., Gardner, J.E., Burgisser, A., **Hajimirza, S.**, Earley, T.C., **Truong, N.** and **Toledo, P.**, Bubble coalescence and percolation threshold in expanding rhyolitic magma. *Geochem. Geophys. Geosys.*, 20, <https://doi.org/10.1029/2018GC008006> (2019).

Moitra, P., Gonnermann, H.M., Houghton B.F. and Tiwary, C.S., Fragmentation and Plinian eruption of crystallizing basaltic magma, *Earth Planet. Sci. Lett.*, 500, 97-104, [doi:10.1016/j.epsl.2018.08.003](https://doi.org/10.1016/j.epsl.2018.08.003) (2018).

Tucker, J.M., Mukhopadhyay, S., Gonnermann, H.M., Reconstructing mantle carbon and noble gas contents from degassed mid-ocean ridge basalts. *Earth Planet. Sci. Lett.*, 496, 108–119, <https://doi.org/10.1016/j.epsl.2018.05.024> (2018).

Jones, M.R., Soule, S.A., Gonnermann, H.M., Le Roux, V., Clague, D.A., Magma ascent and lava flow emplacement rates during the 2011 Axial Seamount eruption based on CO₂ degassing. *Earth Planet. Sci. Lett.*, 494, 32–41, <https://doi.org/10.1016/j.epsl.2018.04.044> (2018).

Gardner, J.E., **Hajimirza, S.**, Webster, J.D., Gonnermann, H.M., The impact of dissolved fluorine on bubble nucleation in hydrous rhyolite melts. *Geochim. Cosmochim. Acta*, 226, 174–181, <https://doi.org/10.1016/j.gca.2018.02.013> (2018).

Liu Z., Dugan, B., Masiello, C.A., **Wahab, L.M.**, Gonnermann, H.M., Nitttrouer, J.A., Effect of freeze-thaw cycling on grain size of biochar. *PLoS ONE*, 13(1): e0191246, doi.org/10.1371/journal.pone.0191246 (2018).

Gonnermann, H.M., **Giachetti, T.**, **Fliedner, C.**, **Nguyen, C.T.**, Houghton, B. F., **Crozier, J.A.** and Carey, R.J., Permeability during magma expansion and compaction. *J. Geophys. Res.*, 122, 1–24, <https://doi.org/10.1002/2017JB014783> (2017).

Simkins, L.M., Anderson, J.B., Greenwood, S.L., Gonnermann, H.M., Prothro, L.O., Halberstadt, A.R.W., Stearns, L.A., Pollard, D. and DeConto, R.M., Anatomy of a meltwater drainage system beneath the ancestral East Antarctic ice sheet, *Nat. Geosci.*, 10, 691-697, [doi:10.1038/ngeo3012](https://doi.org/10.1038/ngeo3012) (2017).

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Gardner, J.E., Jackson, B.A., Gonnermann, H.M. and Soule, S.A., Rapid ascent and emplacement of basaltic lava during the 2005–06 eruption of the East Pacific Rise at ca. 9°51'N as inferred from CO₂ contents, *Earth Planet Sci. Lett.*, 453, 152-160, [doi:10.1016/j.epsl.2016.08.007](https://doi.org/10.1016/j.epsl.2016.08.007), (2016).

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- Giachetti, T.**, Gonnermann, H.M., Gardner, J.E., Shea, T., Gouldstone, A., Discriminating secondary from primary water in rhyolitic matrix-glass of volcanic pyroclasts using thermogravimetric analysis, *Geochim. Cosmochim. Acta*, 148, 457-476, doi:10.1016/j.gca.2014.10.017 (2015).
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Mauna Loa and Kilauea by stress transfer in an asthenospheric melt layer, *Nature Geosci.*, 5, 826-829, doi:10.1038/ngeo1612 (2012).

Gonnermann, H.M. and Houghton, B.F., Magma degassing during the Plinian eruption of Novarupta, Alaska, 1912, *Geophys. Geochem. Geosys.*, 13(10), Q10009, doi:10.1029/2012GC004273 (2012).

Stovall, W.K., Houghton, B.F., Gonnermann, H.M., Fagents, S.A. and Swanson, D.A., Eruption dynamics of Hawaiian-style fountains: the case study of episode 1 of the Kilauea Iki 1959 eruption, *Bull. Volcanol.*, 73:511–529 (2011).

Gonnermann, H.M. and Mukhopadhyay, S., Preserving noble gases in a convecting mantle, *Nature*, 459, 560-563 (2009).

Houghton, B.F. and Gonnermann, H.M., Basaltic explosive volcanism: Constraints from deposits and models, *Chem. Erde Geochem.*, 68, 117-140 (2008).

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Gonnermann, H.M. and Manga, M., The fluid mechanics of volcanic eruptions, *Ann. Rev. Fluid Mech.* 39, 321-356 (2007).

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Gonnermann, H.M., Jellinek, A.M., Richards, M.A. and Manga, M., Modulation of mantle plumes and heat flow at the core mantle boundary by plate-scale flow: results from laboratory experiments, *Earth Planet. Sci. Lett.* 226, 53-67 (2004).

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Gonnermann, H.M., Manga, M. and Jellinek, A.M., Dynamics and longevity of an initially stratified mantle, *Geophys. Res. Lett.* 29, 10.1029/2002GL014851 (2002).

Jellinek, A.M., Gonnermann, H.M. and Richards, M.A., Plume capture by divergent plate motions: Implications for the distribution of hotspots, geochemistry of mid-ocean ridge basalts, and estimates of the heat flux at the core-mantle boundary, *Earth Planet. Sci. Lett.* 205, 361-378 (2002).

Mok, C.M., Chang, C.-Y., Settgest, R., Wang, Z.L., Gonnermann, H.M., Chin, C.C., Dynamic soil foundation structure interaction analysis of large caissons, *U.S. Geol. Survey Open-File Report* 99-0142 19.1-19.30 (1999).

Invited Papers

Gonnermann, H.M., Magma fragmentation, *Annu. Rev. Earth Planet. Sci.*, 43, doi:10.1146/annurev-earth-060614-105206 (2015).

Gonnermann, H.M. and Taisne, B., Magma Transport in Dikes, in 2nd edition of the *Encyclopedia of Volcanoes* (2015).

Gonnermann, H.M. and Manga, M., Dynamics of magma ascent in the volcanic conduit, in *Modeling Volcanic Processes, The Physics and Mathematics of Volcanism*, eds. Fagents, S.A., Gregg, T.K.P. and Lopes, R.M.C., Cambridge University Press, Cambridge (2013).

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